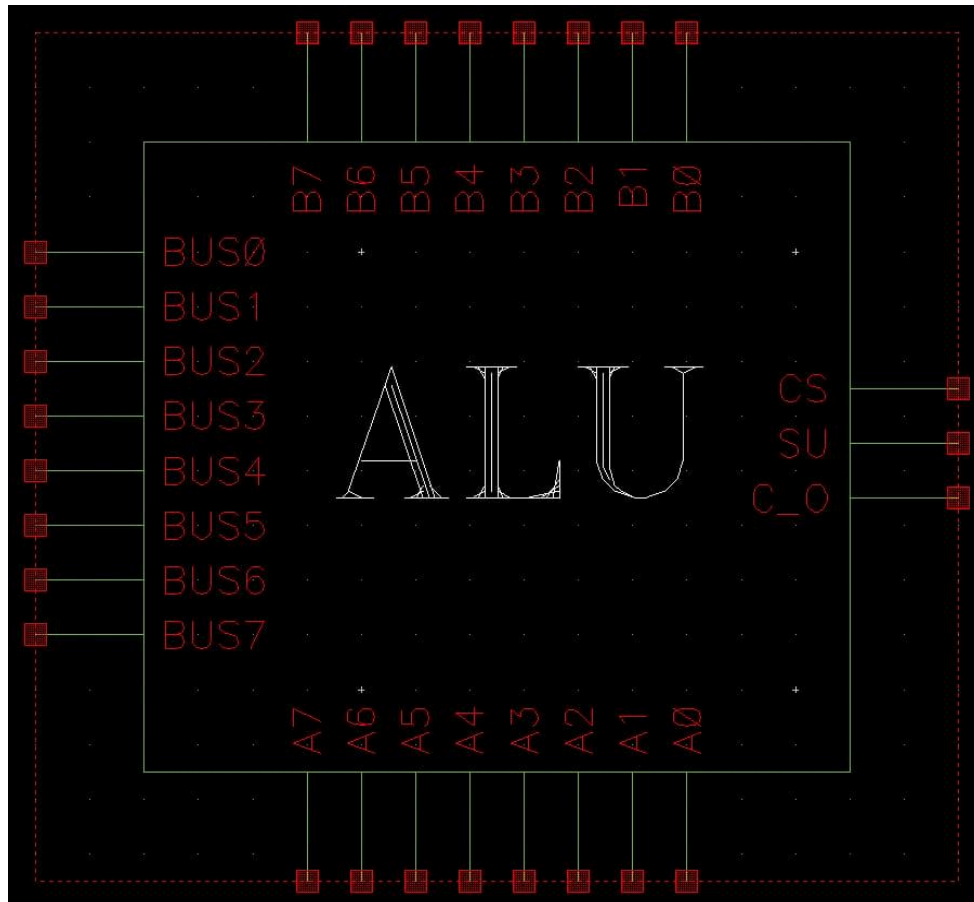


ASIC Design of a 8-Bit ALU

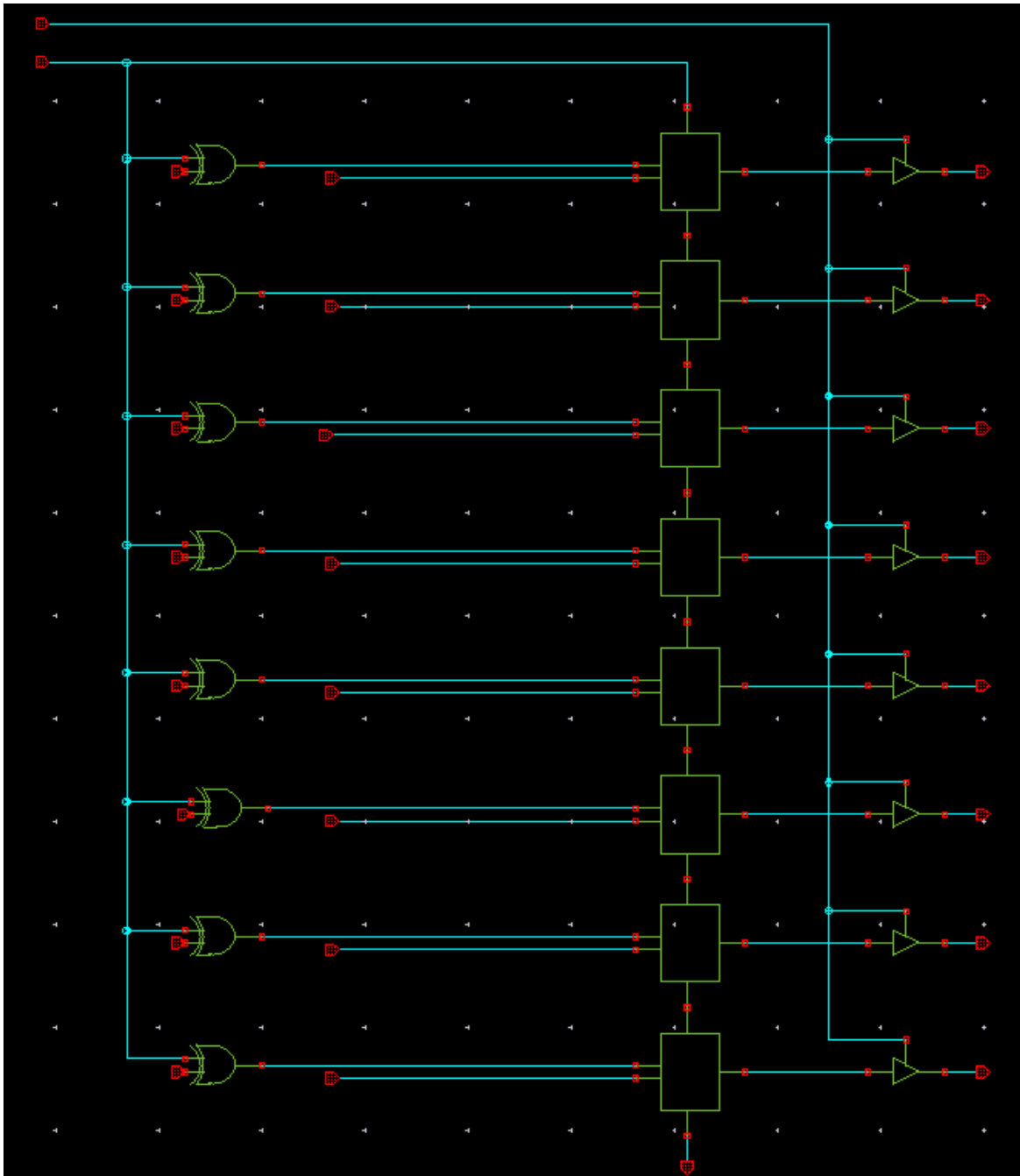


Team :

- Het Patel : 202304033
- Devansh Modi : 202304009
- Malay Vaghasiya : 202304027

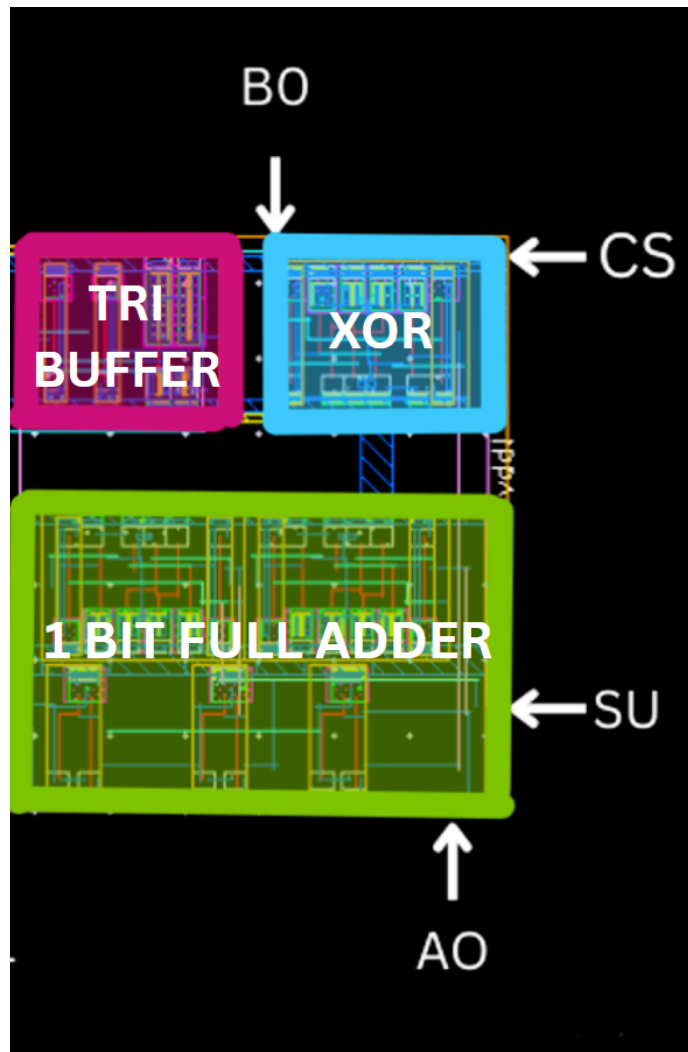
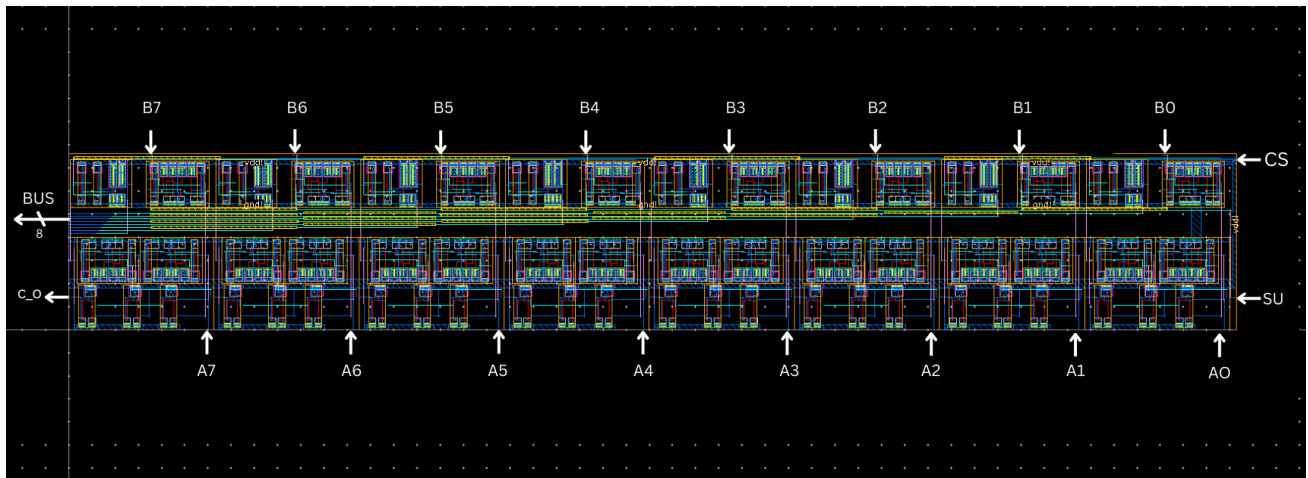
I/O:

- **Inputs** : (A7 - A0), (B7 - B0), CS, SU
- **Outputs** : (BUS7 - BUS0), C_O



- Input from register A and register B is **hard wired** to the ALU.
- Input from register B and SU is connected to the XOR Gate, and the output of the XOR gate is inserted as 2nd input of the ALU.

Layout:



Functionality :

- **A0 - A7** and **B0 - B7** are input pins hard wired to register A and register B, respectively.
- A0 & B0 are LSB and A7 & B7 are MSB.
- Pins (**BUS0 - BUS7**) contain the output of the ALU, which will be connected to the bus.
- **C_O** is the output carry.
- The **SU** pin is used to switch between addition and subtraction. It is also connected to the input carry.
 - **SU = 0**. Addition is performed by the ALU.
 - **SU = 1**. Subtraction is performed by the ALU.
- **CS** is the chip select pin in an active high configuration.
 - **CS = 1**, Output of ALU will be connected to the bus.
 - **CS = 0**, Output will be in high impedance state (i.e. ALU will be isolated).

Timings and Delay :

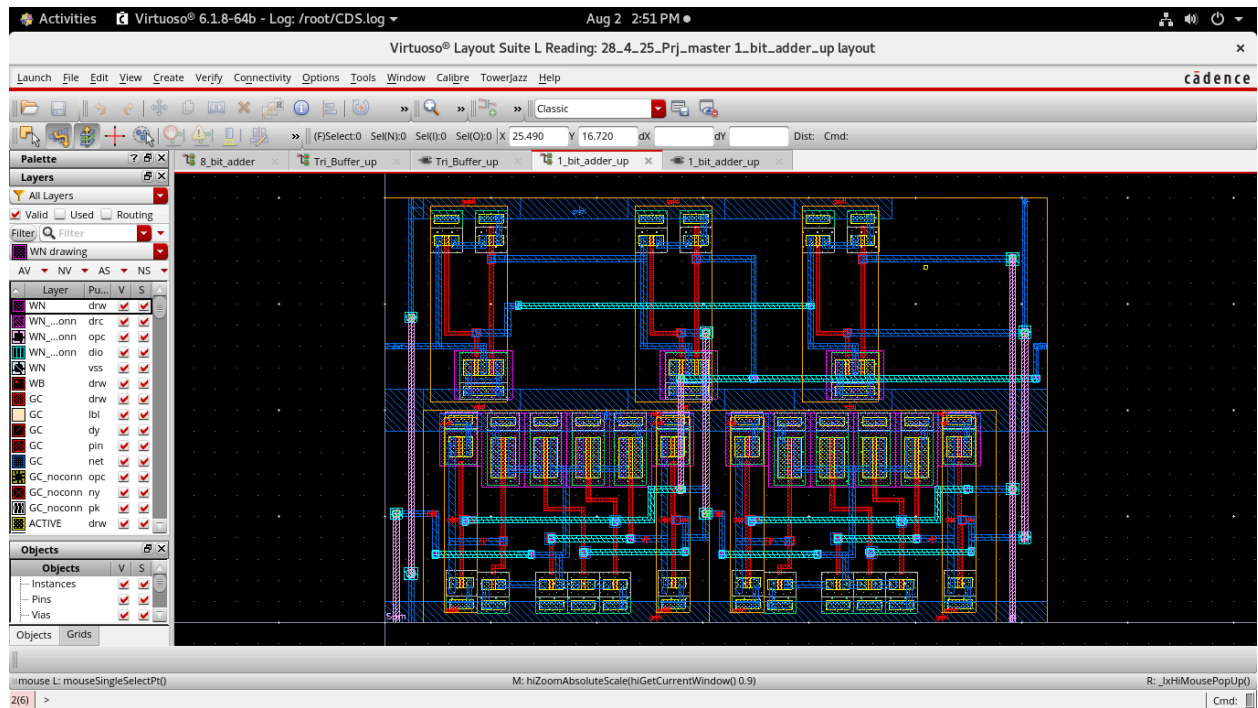
- After performing post-layout simulation, the **propagation delay of the ALU** was found to be **1.62 ns**.
- The maximum current ALU can provide is 535 μ A, and it can charge a 50 pf capacitor in 0.2 μ s.

Tri-State Buffer Implementation :

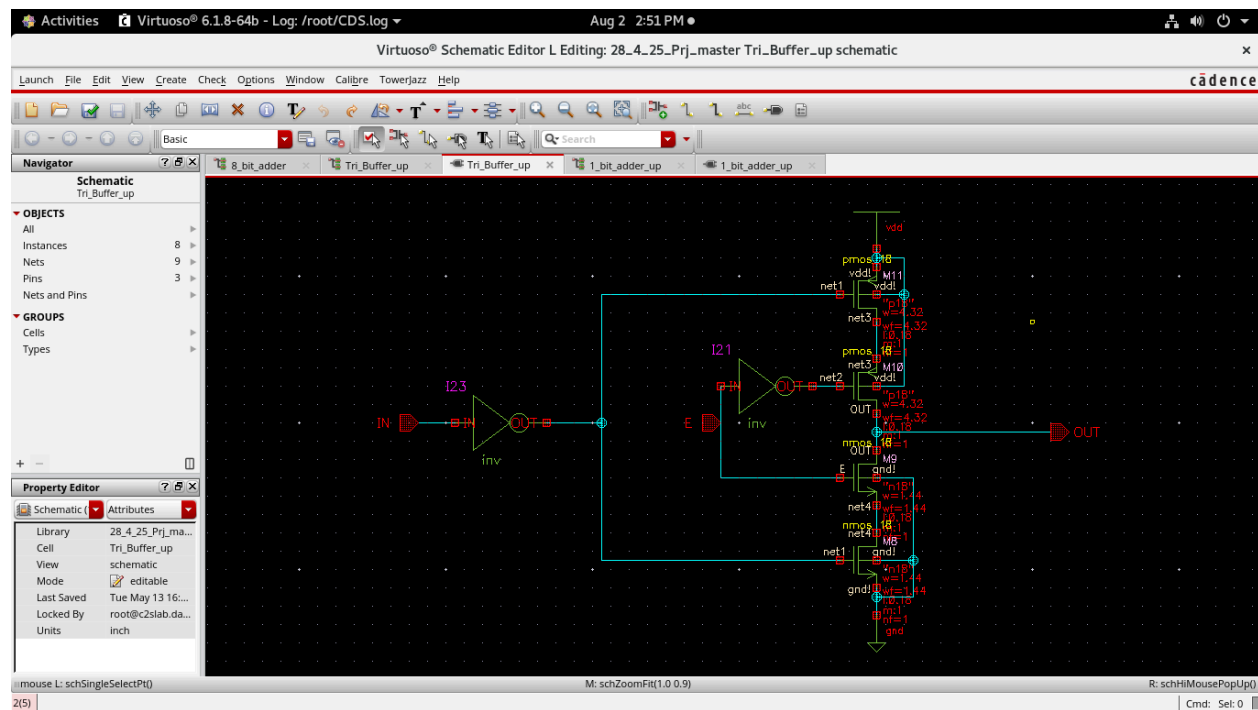
To ensure the ALU can be safely integrated into a larger system architecture containing a shared data bus, the output stage utilises tri-state buffers controlled by the Chip Select (CS) pin.

- Active State (CS = 1): The buffer turns on, actively driving the calculated results from the 1-bit full adders onto the output pins (BUS0 - BUS7).

High-Impedance State (CS = 0): The buffer is disabled, isolating the ALU from the bus. This prevents electrical contention (short circuits) when another component attempts to write to the shared bus simultaneously.



Tristate Buffer Layout



Tristate Buffer schematic

Worst-Case Delay Analysis:

The propagation delay of 1.62 ns represents the critical path of the ALU. The worst-case timing scenario occurs during an operation that forces a carry bit to ripple through every single stage of the 8-bit architecture.

The Test Vector:

This worst-case delay is triggered by applying the input vectors **A = 00000001** and **B = 11111111** during an addition operation (SU=0).

- Adding Bit 0 generates a carry ($1 + 1 = 10$).
- This carry must propagate through Bit 1, generating another carry that cascades all the way to Bit 7.

The Critical Path:

The exact physical signal path for this maximum delay is as follows:

1. Input arrival at A0 / B0.
2. Propagation through the input XOR gate.
3. Sequential propagation through the carry chains of the 1-bit Full Adders (Bit 0 through Bit 7).
4. Propagation of the final Bit 7 sum through the Tri-state buffer.
5. Arrival at the final output pin, BUS7 (Sout7).

The CS Pin: The CS pin can also act as the worst-case delay input if it is asserted late. If the mathematical addition completes before the CS pin is driven high, the delay is determined solely by the turn-on time of the output tri-state buffers driving the bus load, rather than by the internal ripple-carry logic.

Data Range, Maximum Limits, and Overflow

Because this is an 8-bit architecture, the maximum representable data range depends entirely on whether the system interprets the bits as unsigned or signed (2's complement) numbers.

- Unsigned Range: 0 to 255.
- Signed Range (2's Complement): **-128 to 127**.

The Failure of C_{out} in Signed Arithmetic:

When performing unsigned arithmetic, the final carry-out pin C_out perfectly flags an overflow (e.g., exceeding 255). However, when dealing with signed numbers, C_out is not a reliable indicator of mathematical overflow.

Example: Adding -2 and -4

- -2: 1111 1110
- -4: 1111 1100
- Sum: 1 1111 1010

In this operation, a carry out (1) is generated at the Most Significant Bit (MSB). However, the resulting 8-bit value is 1111 1010, which equals -6. Because -6 is well within the valid -128 to 127 range, no actual overflow has occurred. Relying strictly on C_out would yield a false error.

Design Limitations and the XOR Solution:

A current limitation of this ALU design is the absence of a dedicated Signed Overflow Flag V. While it outputs C_out for unsigned overflow, signed overflow remains undetected by the hardware.

Signed overflow occurs exclusively when the sum of two numbers with the same sign yields a result with the opposite sign (e.g., Positive + Positive = Negative).

The XOR Solution:

To resolve this, the design requires an XOR gate comparing the carry entering the MSB to the carry leaving the MSB:

$$V = C_{in7} \text{ XOR } C_{out7}$$

Example of the XOR Logic in Action (64 + 64):

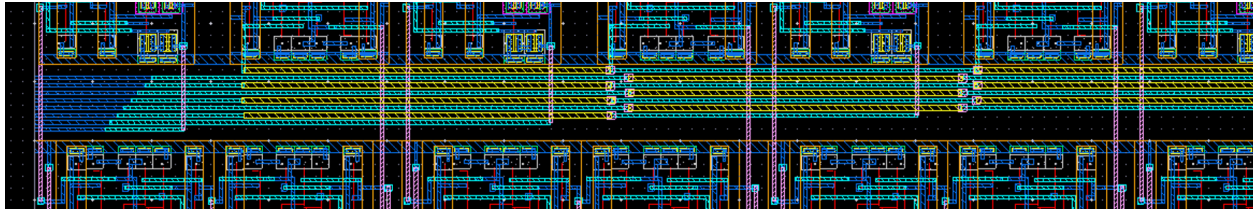
If the ALU adds 64 and 64, the result (128) exceeds the signed maximum limit of 127.

- 64: 0100 0000
- 64: 0100 0000
- Sum: 1000 0000 (Interpreted as -128 in 2's complement).

Here is how the XOR logic correctly flags this catastrophic failure:

1. A carry ripples from Bit 6 into Bit 7, so the carry-in to the MSB C_in7 is 1.
2. Bit 7 calculates $0 + 0 + 1 = 1$, generating no carry-out, so C_out7 is 0.
3. The XOR evaluates $1 \text{ xor } 0 = 1$. The system correctly flags a signed overflow, preventing the CPU from interpreting the result as -128.

5. Antenna Effect and Layout Prevention Strategies



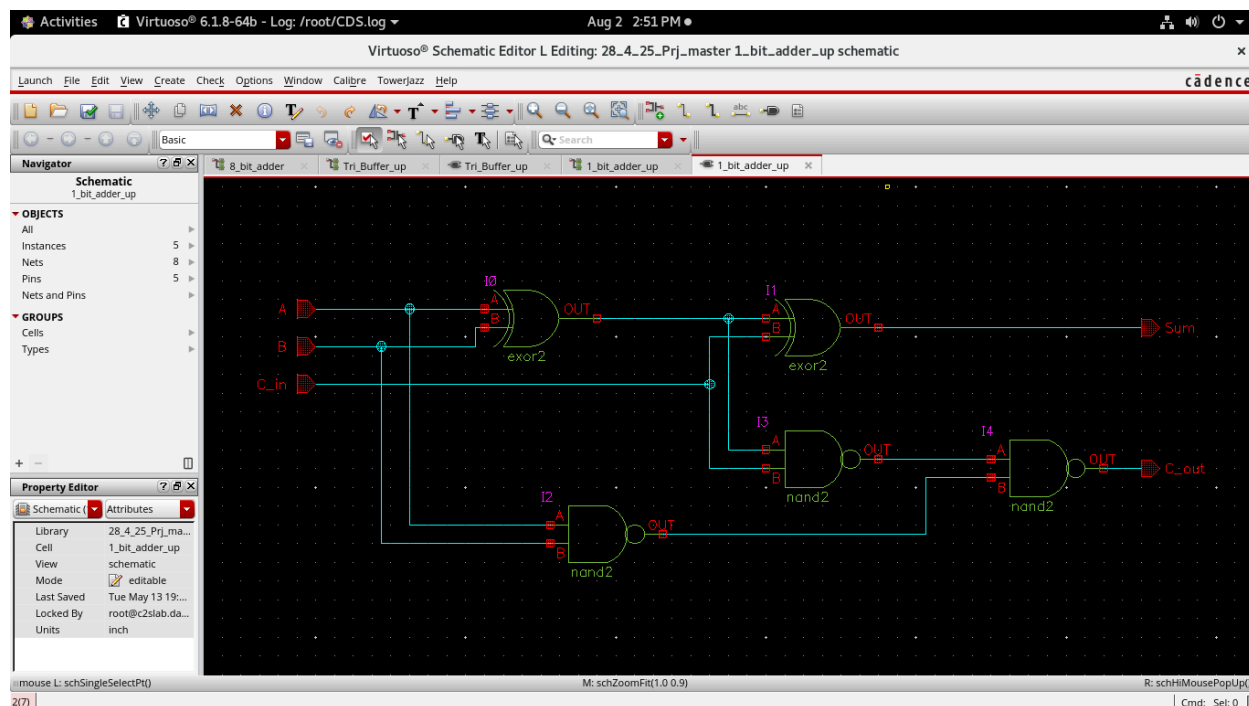
During fabrication, particularly during plasma etching, long, unbroken floating metal traces can act as antennas. These traces collect massive amounts of static charge from the plasma. If this accumulated charge exceeds the breakdown voltage of the thin gate oxide of the transistor it drives, it will violently discharge and permanently destroy the device.

Metal Jumpering (Layer Hopping) Technique:

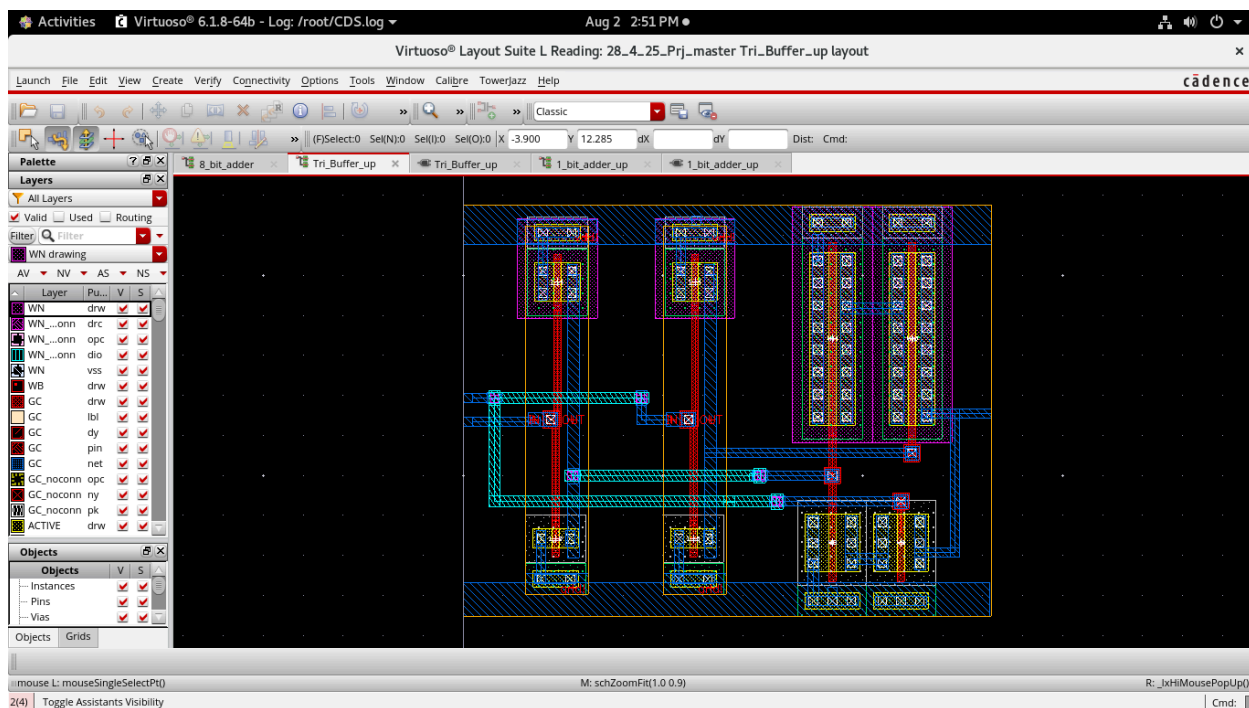
To resolve "Antenna ratio exceeded" violations and ensure manufacturability, the layout employs a "metal jumpering" routing strategy.

- As demonstrated in the central routing channel of the layout, long horizontal data lines are not drawn as a single continuous piece of metal on one layer.
- Instead, the long traces are deliberately severed into shorter segments by dropping down (or popping up) to an adjacent metal layer using vias for a short distance, before returning to the primary routing layer.
- Because the Damascene fabrication process etches the wafer layer-by-layer, breaking the trace ensures that no single metal layer is physically long enough to accumulate a lethal amount of plasma charge during its specific etching step. The temporary discontinuity safely isolates the sensitive transistor gates until the final upper-level metal layers bridge the connections.

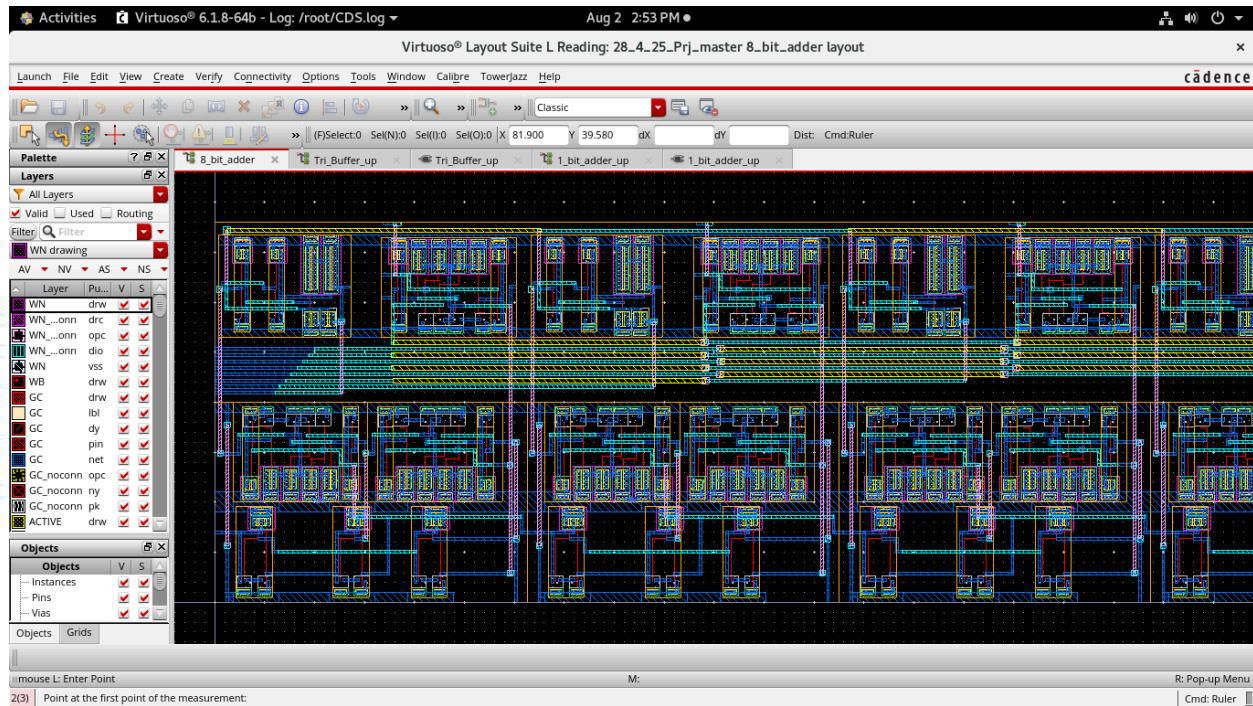
Images:



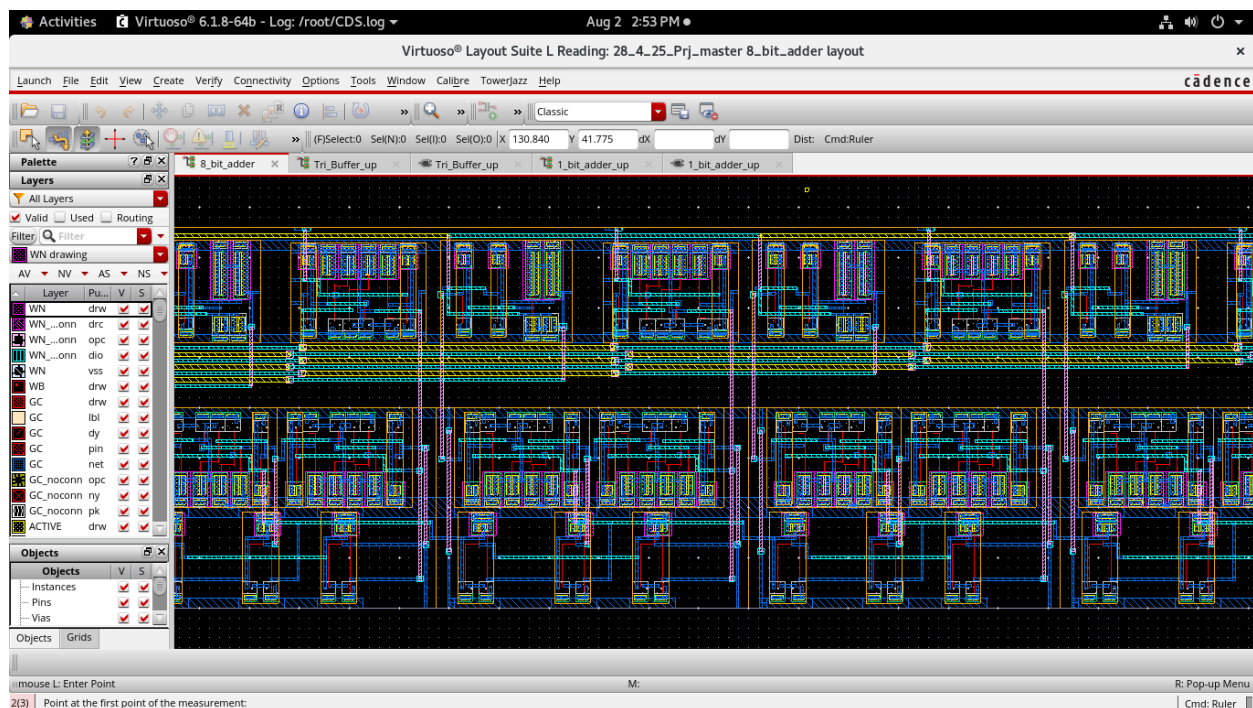
Full adder schematic



Full adder Layout:



Zoomed-in version of the ALU layout



Zoomed-in version of the ALU layout